

Report

Of

Student Registration System

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)

By

Roll Number of Student: [Your Roll Number]

Name of Student: SALONI KUMARI

Batch: 2025-2029

CANDIDATE'S DECLARATION

I, SALONI KUMARI, do hereby declare that the internship report titled “Student Registration System” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: SALONI KUMARI

Roll No.: [25scs1003002717]

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: SALONI KUMARI

Roll No.: 25scs1003002717

TABLE OF CONTENTS

Chapter No. / Chapter Name	Page Nos.
Cover Page	----
1. Candidate's Declaration	i
2. Acknowledgment	ii
3. Internship Completion Certificate	----
4. Project Description	
4.1 Introduction	1
4.2 Organization Profile	1
4.3 Problem Statement	2
4.4 Project Objectives	2-3
4.5 Scope of the Project	3
4.6 Technologies and Tools Used	3-5
4.7 System Architecture	5-6
4.8 Methodology	7
4.9 Expected Outcomes	7-8
4.10 Certificates of completion and other communication mail proof must be attached	8-9
5. Bibliography/References	10

3. Internship Completion Certificate



Chapter 4: Project Description

4.1 Introduction

The Student Registration System is a full-stack web application developed to simplify and digitize the process of registering and managing student records for an educational institution. The application

allows administrators to add, view, update, and delete student records through a secure and easy-to-use interface, replacing manual, paper-based registration processes.

Users can register students by entering their personal and academic details, and the system stores this information in a centralized database that can be searched and updated as required. Secure login ensures that only authorized users can access and manage student data.

This project was developed as an Internship Capstone Project during the Lenovo LEAP NextGen Scholar Program, a 6-week AICTE internship conducted during June–July 2026 in association with the All India Council for Technical Education (AICTE) and implemented by BharatCares. The project aims to make student record management more organized, accurate, and accessible. It is aligned with United Nations Sustainable Development Goal (SDG) 4 – Quality Education, which promotes inclusive and equitable quality education and lifelong learning opportunities for all.

4.2 Organization Profile

The internship was conducted under the Lenovo LEAP NextGen Scholar Program, a CSR initiative of Lenovo India, in association with AICTE (All India Council for Technical Education) and implemented by BharatCares (SMEC Trust). The program is designed to provide undergraduate students with industry-relevant skills and hands-on experience in emerging technology domains.

Lenovo Group Limited is a Fortune Global 500 multinational technology company headquartered in Beijing, China, and Morrisville, North Carolina, USA. Lenovo designs, develops, manufactures, and sells personal computers, tablets, smartphones, workstations, servers, electronic storage devices, IT management software, and smart televisions. Lenovo is one of the world's largest personal computer vendors by unit sales.

4.3 Problem Statement

Many educational institutions still rely on manual or paper-based methods for registering and maintaining student records, which leads to several challenges:

- Student data is often scattered across registers and files, making it difficult to search or update.
- Manual record-keeping is prone to human error, duplication, and loss of data.
- Retrieving or updating a student's information takes considerable time and effort.
- There is no centralized, secure system to manage student records digitally.

Core Problem: How can a web-based system help institutions register students, store their information securely, and manage student records efficiently through a centralized platform?

4.4 Project Objectives

The main objectives of the Student Registration System are:

- Develop a web-based platform that allows administrators to register new students with their personal and academic details.
- Provide functionality to view, update, and delete existing student records.
- Implement secure user authentication using JWT authentication and bcrypt password hashing.
- Store and manage student data reliably using MongoDB Atlas and Mongoose.
- Expose RESTful APIs to handle communication between the frontend and backend.

- Build a responsive and user-friendly interface using React.js, Vite, HTML5, CSS3, and JavaScript.

4.5 Scope of the Project

The scope of the Student Registration System includes the following functionalities:

In Scope:

- Secure admin login and authentication.
- Registration of new students with personal and academic details.
- Viewing, updating, and deleting student records.
- Searching and filtering student records.
- Responsive web interface for desktop and mobile devices.

Out of Scope (Future Enhancements):

- Native Android/iOS mobile application.
- Fee management and payment integration.
- Attendance tracking module.
- Automated email/SMS notifications to students.
- Advanced analytics and reporting dashboards.

4.6 Technologies and Tools Used

The following technologies, frameworks, libraries, and tools were used in the development of the Student Registration System:

Frontend Technologies:

Technology	Purpose
HTML5	Structure and semantic markup
CSS3	Styling, layout, and responsive design
JavaScript (JSX)	Client-side logic and application functionality
React.js	Building interactive user interfaces
Vite	Frontend development and build tool
React Router	Client-side navigation
Axios	API communication between frontend and backend

Backend Technologies:

Technology	Purpose
------------	---------

Node.js	Server-side JavaScript runtime
Express.js	Backend framework and REST API development
JWT (jsonwebtoken)	User authentication
bcryptjs	Secure password hashing
dotenv	Environment variable management

Database:

Technology	Purpose
MongoDB Atlas	Cloud database for storing application data
Mongoose	ODM for schemas, models, and MongoDB operations

Development & Deployment Tools:

Tool	Purpose
Visual Studio Code	Code editor and development environment
Git & GitHub	Version control and repository management
Vercel	Frontend deployment
Render	Backend deployment
Postman	API testing

4.7 System Architecture

The Student Registration System follows a three-tier client-server architecture with a clear separation between the frontend, backend, and data layers. The frontend communicates with the backend through RESTful APIs, while the backend manages business logic, authentication, and database operations.

Architecture Overview:

- Presentation Layer (Frontend):** The frontend is developed using React.js, Vite, HTML5, CSS3, and JavaScript. It provides the user interface for admin login, student registration, viewing records, and searching/updating student data. React Router is used for navigation, while Axios is used for communication with the backend REST APIs.
- Application Layer (Backend):** The backend is implemented using Node.js and Express.js. It handles request routing, business logic, authentication, validation, and communication between the frontend and the database. The backend exposes RESTful API endpoints that are consumed by the frontend.
- Data Layer:** MongoDB Atlas is used as the cloud database for storing student and user data, while Mongoose is used for defining schemas and performing database operations.

Project Directory Structure:

```
StudentRegistrationSystem/ | frontend/ | public/ | src/ | components/
| | pages/ | routes/ | utils/ | App.jsx | main.jsx
| backend/ | config/ | controllers/ | middleware/ | models/ |
| routes/ | server.js | README.md
```

Data Flow:

User → React Frontend → Axios/REST API → Express Backend → Mongoose → MongoDB Atlas
→ Response → Frontend → User

Database Schema:

The Student Registration System uses MongoDB Atlas as its database, with Mongoose used for schema definition and database management. The application contains separate collections/models for managing users and student data.

Collection/Model	Purpose
User	Stores admin account and authentication information
Student	Stores student personal and academic details

4.8 Methodology

The development of the Student Registration System followed an Agile Software Development Methodology with iterative development and testing cycles. The project was completed through the following phases:

- Requirement Analysis – Identified the challenges of manual student record-keeping and defined the requirements for a digital registration system.
- System Planning – Analyzed the project requirements and planned the application features, technology stack, and overall workflow.
- UI & Database Design – Designed the user interface and defined MongoDB/Mongoose schemas for users and student records.
- Backend Development – Developed the Node.js and Express.js backend with RESTful APIs for handling registration and record management.
- Authentication & Security – Implemented JWT-based authentication, bcrypt password hashing, and validation for secure access.
- API Integration – Connected the frontend with backend REST APIs using Axios.
- Frontend Development – Developed responsive and interactive interfaces using React.js, Vite, HTML5, CSS3, and JavaScript.
- Feature Development – Implemented student registration, record viewing/updating/deleting, and search functionality.
- Integration & Testing – Connected the frontend, backend, and database, tested APIs and application features, and fixed identified issues.
- Deployment & Documentation – Deployed the application, maintained the source code using Git and GitHub, and prepared project documentation.

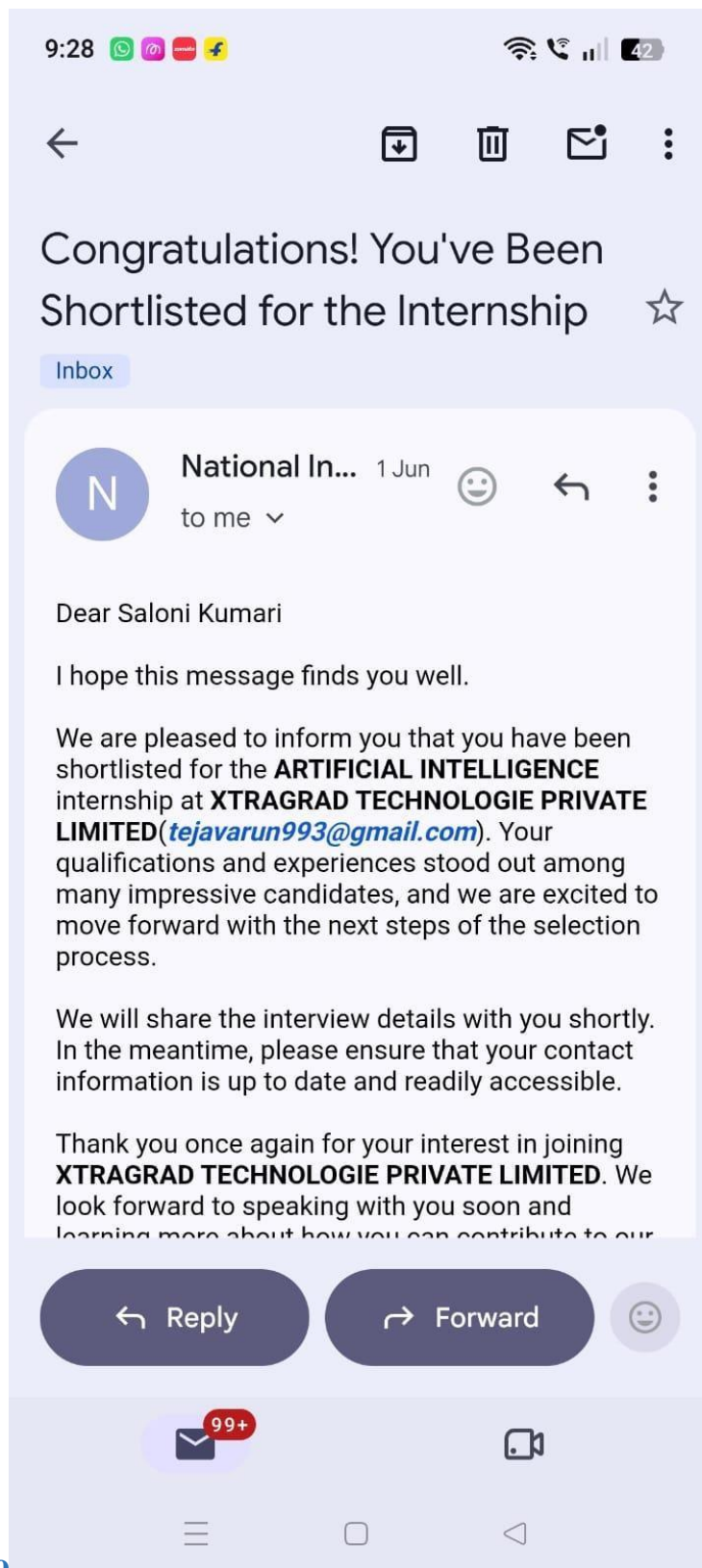
4.9 Expected Outcomes

The expected outcomes of the Student Registration System project are:

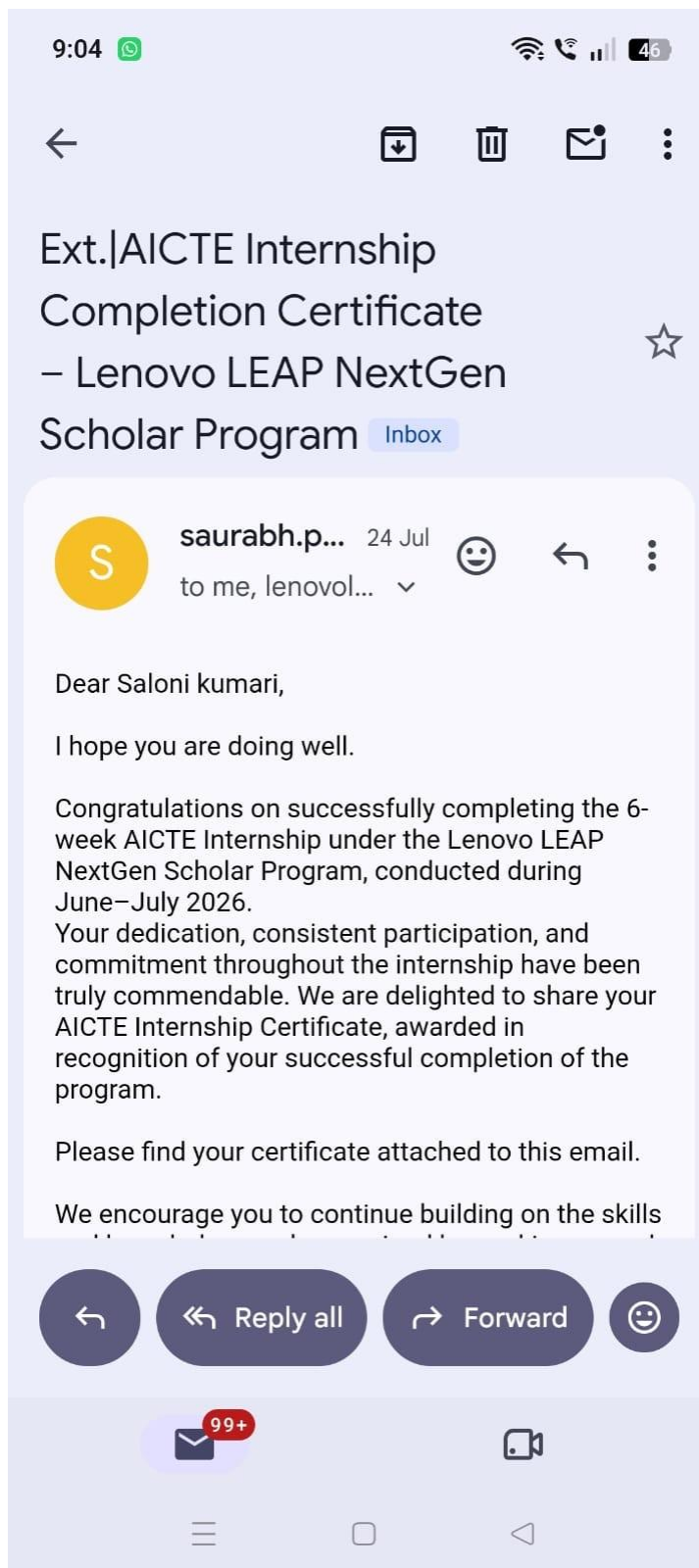
- A functional Student Registration System that allows administrators to register and manage student records digitally.
- Secure user authentication and data management using JWT, bcrypt, REST APIs, MongoDB Atlas, and Mongoose.
- An organized, centralized database of student records that is easy to search, update, and maintain.
- A responsive and user-friendly interface accessible on both desktop and mobile devices.
- Practical full-stack implementation using React.js, Vite, Node.js, Express.js, HTML5, CSS3, JavaScript, and MongoDB.

4.10 Certificates of Completion and Communication Proof

- Internship Completion Certificate issued by Lenovo LEAP NextGen Scholar Program (attach on the following page).
- AICTE internship completion acknowledgment.
- Communication emails and correspondence with mentors during the internship period.
- GitHub repository link: [Your GitHub repository link]
- Live link: [Your deployed project link]



C0mmunication Proo



Chapter 5: Bibliography/References

[1] React, “React Documentation,” <https://react.dev/>.

- [2] Vite, “Vite Documentation,” <https://vite.dev/>.
- [3] Node.js, “Node.js Documentation,” <https://nodejs.org/docs/latest/>.
- [4] Express.js, “Express.js Documentation,” <https://expressjs.com/>.
- [5] MongoDB, “MongoDB Documentation,” <https://www.mongodb.com/docs/>.
- [6] Mongoose, “Mongoose Documentation,” <https://mongoosejs.com/docs/>.
- [7] Lenovo LEAP: NextGen Scholar Program, “Lenovo LEAP – NextGen Scholar Program,” <https://leapnextgenscholar.in/>.
- [8] All India Council for Technical Education (AICTE), “Lenovo LEAP: Student Registration System Internship,” AICTE Internship Portal.